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## 14. SQL Window Functions

*Compiled by Bikkad IT Institute | Teaching Notes*

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# SQL Window Functions

## 14 · Window Functions Basics — Complete Reference Guide

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## 1. Window Functions — Basics

→ *Task 1: Total sales across all orders (standard aggregate)*

```
SELECT SUM(Sales) AS Total_Sales  
FROM Sales.Orders;
```

→ *Task 2: Total sales for each product (standard GROUP BY)*

```
SELECT  
    ProductID,  
    SUM(Sales) AS Total_Sales  
FROM Sales.Orders  
GROUP BY ProductID;
```

## 2. OVER Clause — Window Aggregate

→ *Task 3: Total sales across all orders — keep row-level detail*

```
SELECT  
    OrderID,  
    OrderDate,  
    ProductID,
```

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```
Sales,
SUM(Sales) OVER () AS Total_Sales
FROM Sales.Orders;
```

### 3. PARTITION Clause

→ *Task 4: Total sales overall AND per product*

```
SELECT
    OrderID, OrderDate, ProductID, Sales,
    SUM(Sales) OVER () AS Total_Sales,
    SUM(Sales) OVER (PARTITION BY ProductID) AS Sales_By_Product
FROM Sales.Orders;
```

→ *Task 5: Total sales overall, per product, and per product+status*

```
SELECT
    OrderID, OrderDate, ProductID, OrderStatus, Sales,
    SUM(Sales) OVER () AS Total_Sales,
    SUM(Sales) OVER (PARTITION BY ProductID) AS Sales_By_Product,
    SUM(Sales) OVER (PARTITION BY ProductID, OrderStatus) AS Sales_By_Product_Status
FROM Sales.Orders;
```

### 4. ORDER Clause — Ranking

→ *Task 6: Rank each order by Sales from highest to lowest*

```
SELECT
    OrderID, OrderDate, Sales,
    RANK() OVER (ORDER BY Sales DESC) AS Rank_Sales
FROM Sales.Orders;
```

### 5. FRAME Clause

→ *Task 7: Running total — current row and next 2 rows*

```
SELECT
    OrderID, OrderDate, OrderStatus, Sales,
    SUM(Sales) OVER (
        PARTITION BY OrderStatus
        ORDER BY OrderDate
        ROWS BETWEEN CURRENT ROW AND 2 FOLLOWING
    ) AS Total_Sales
FROM Sales.Orders;
```

→ *Task 8: Rolling total — previous 2 rows and current row*

```
SELECT
    OrderID, OrderDate, OrderStatus, Sales,
    SUM(Sales) OVER (
```

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```

    PARTITION BY OrderStatus
    ORDER BY OrderDate
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
  ) AS Total_Sales
FROM Sales.Orders;

```

→ *Task 9: Previous 2 rows only*

```

SELECT
  OrderID, OrderDate, OrderStatus, Sales,
  SUM(Sales) OVER (
    PARTITION BY OrderStatus
    ORDER BY OrderDate
    ROWS 2 PRECEDING
  ) AS Total_Sales
FROM Sales.Orders;

```

→ *Task 10 & 11: Cumulative total from start to current row*

```

SELECT
  OrderID, OrderDate, OrderStatus, Sales,
  SUM(Sales) OVER (
    PARTITION BY OrderStatus
    ORDER BY OrderDate
    ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
  ) AS Total_Sales
FROM Sales.Orders;

```

## 6. Window Function Rules

Rule 1 — Window functions can ONLY be used in SELECT or ORDER BY clauses.

They CANNOT be used in WHERE, HAVING, or GROUP BY.

```

-- INVALID: Window function in WHERE clause
SELECT OrderID, Sales,
  SUM(Sales) OVER (PARTITION BY OrderStatus) AS Total_Sales
FROM Sales.Orders
WHERE SUM(Sales) OVER (PARTITION BY OrderStatus) > 100; -- ERROR

```

Rule 2 — Window functions CANNOT be nested inside other window functions.

```

-- INVALID: Nested window functions
SELECT
  SUM(SUM(Sales) OVER (PARTITION BY OrderStatus))
  OVER (PARTITION BY OrderStatus) AS Total_Sales -- ERROR
FROM Sales.Orders;

```

## 7. Window Functions with GROUP BY

→ *Task 12: Rank customers by their total sales*

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```
SELECT
    CustomerID,
    SUM(Sales) AS Total_Sales,
    RANK() OVER (ORDER BY SUM(Sales) DESC) AS Rank_Customers
FROM Sales.Orders
GROUP BY CustomerID;
```

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*SQL Window Functions Guide · Topic 14 · Basics*

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